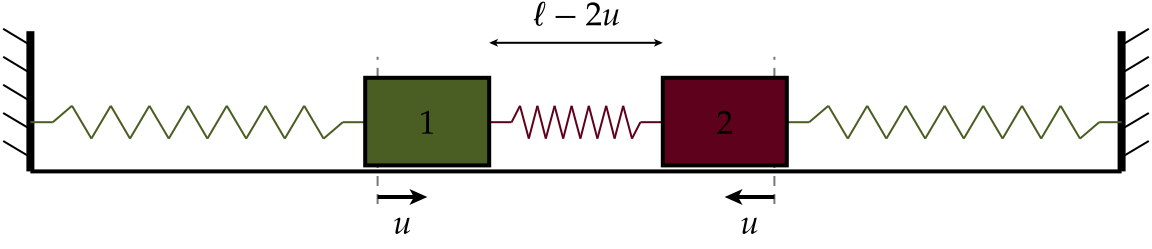
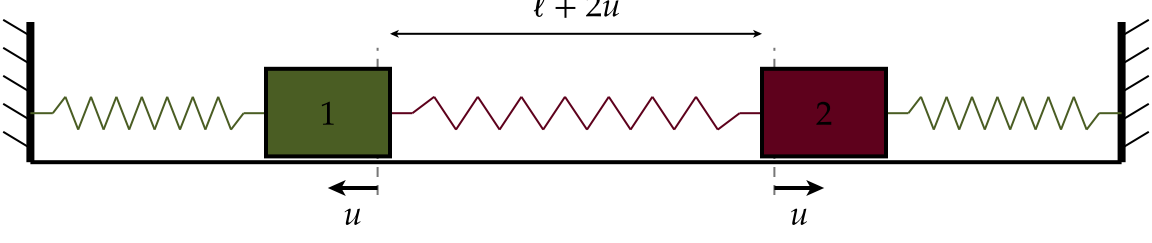
(a) The carts move toward each other: $u_1 = u,\ u_2 = -u$



(b) Half a period later: they move apart